

Enterprise Internetworking Project

Cisco Packet Tracer Network Simulation

OSPF + EIGRP | Dual-Stack IPv4/IPv6 | VTP | STP | DHCP

Craig Ssemakula

28th November, 2025

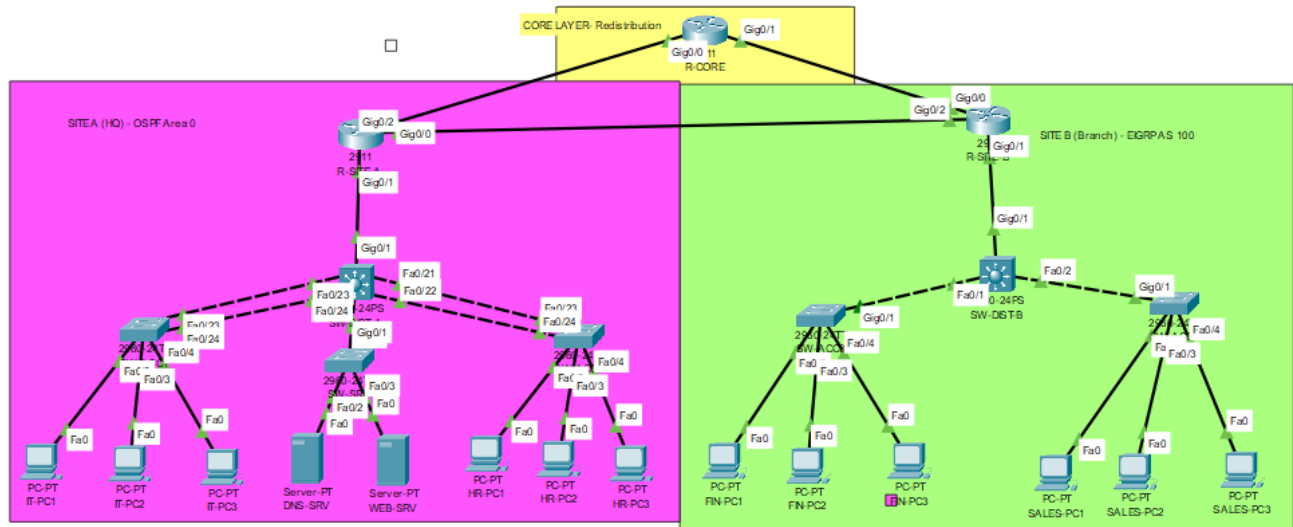
Internetworking

1. Project Overview

This project demonstrates a complex enterprise-level network simulation integrating routing, switching, VLAN management, and advanced internetworking concepts. The network connects two sites (HQ and Branch) using dual dynamic routing protocols with full IPv4/IPv6 dual-stack implementation.

Network Summary

Component	Details
Routers	3 (R-CORE, R-SITE-A, R-SITE-B) - Cisco 2911
Switches	7 (2 Distribution 3560, 5 Access 2960)
End Devices	12 PCs + 2 Servers (DNS, Web)
VLANs	6 (Mgmt, IT, HR, Finance, Servers, Sales)
Dynamic Routing	OSPF Area 0 + EIGRP AS 100 with redistribution
IP Addressing	Dual-stack: IPv4 (10.x.x.x) + IPv6 (2001:db8::/64)



2. Topology and Address Tables

2.1 VLAN Addressing Scheme

The network uses a hierarchical IP addressing scheme with /24 subnets for each VLAN, providing 254 usable host addresses per department. IPv6 uses /64 prefixes following the 2001:db8:: documentation range.

VLAN	Name	IPv4 Network	IPv6 Network	Gateway
10	Management	10.10.10.0/24	2001:db8:10::/64	.1
20	IT_Dept	10.10.20.0/24	2001:db8:20::/64	.1
30	HR_Dept	10.10.30.0/24	2001:db8:30::/64	.1
40	Finance	10.10.40.0/24	2001:db8:40::/64	.1
50	Servers	10.10.50.0/24	2001:db8:50::/64	.1
60	Sales	10.10.60.0/24	2001:db8:60::/64	.1

2.2 Router Point-to-Point Links

Link	Interface	IPv4	IPv6
R-CORE to R-SITE-A	G0/0 - G0/0	10.0.12.0/30	2001:db8:12::/64
R-CORE to R-SITE-B	G0/1 - G0/0	10.0.13.0/30	2001:db8:13::/64
R-SITE-A to R-SITE-B	G0/2 - G0/2	10.0.23.0/30	2001:db8:23::/64

3. Routing Implementation

3.1 Static Routing

Static routes were configured as backup paths between R-SITE-A and R-SITE-B with an administrative distance of 200. These routes activate only if the primary dynamic routing paths fail, providing redundancy for inter-site communication without relying solely on dynamic protocols.

R-SITE-A Static Routes:

- ip route 10.10.40.0 255.255.255.0 10.0.23.2 200 (Finance via backup)
- ip route 10.10.60.0 255.255.255.0 10.0.23.2 200 (Sales via backup)

R-SITE-B Static Routes:

- ip route 10.10.20.0 255.255.255.0 10.0.23.1 200 (IT via backup)
- ip route 10.10.30.0 255.255.255.0 10.0.23.1 200 (HR via backup)

3.2 Dynamic Routing - OSPF

OSPF Area 0 was implemented between R-CORE and R-SITE-A to handle Site A's network advertisements. OSPF was chosen for its fast convergence and support for VLSM. R-SITE-A advertises VLANs 10, 20, 30, and 50 into OSPF with passive interfaces on LAN-facing subinterfaces to prevent unnecessary hello packets on end-user networks.

```
R-CORE#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/DR	00:00:33	10.0.12.2	GigabitEthernet0/0

3.3 Dynamic Routing - EIGRP

EIGRP AS 100 was configured between R-CORE and R-SITE-B for Site B's networks. EIGRP provides fast convergence with DUAL algorithm and efficient bandwidth usage with partial updates. R-SITE-B advertises VLANs 40 and 60 into EIGRP, with passive interfaces preventing EIGRP relationships on LAN segments.

```
R-SITE-B#show ip eigrp neighbors
```

```
IP-EIGRP neighbors for process 100
```

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
0	10.0.13.1	Gig0/0	13	00:33:12	40	1000	0	77

3.4 Route Redistribution

R-CORE performs mutual redistribution between OSPF and EIGRP, allowing Site A networks to reach Site B and vice versa. OSPF routes appear as EIGRP external (D EX) routes on R-SITE-B, and EIGRP routes appear as OSPF external type 2 (O E2) routes on R-SITE-A. This demonstrates proper network convergence across different routing domains.

```
R-CORE#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
```

```
* - candidate default, U - per-user static route, o - ODR
```

```
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
```

```
C 10.0.12.0/30 is directly connected, GigabitEthernet0/0
```

```
L 10.0.12.1/32 is directly connected, GigabitEthernet0/0
```

```
C 10.0.13.0/30 is directly connected, GigabitEthernet0/1
```

```
L 10.0.13.1/32 is directly connected, GigabitEthernet0/1
```

```
D 10.10.10.0/24 [90/28416] via 10.0.13.2, 00:34:54, GigabitEthernet0/1
```

```
O 10.10.20.0/24 [110/2] via 10.0.12.2, 04:15:05, GigabitEthernet0/0
```

```
O 10.10.30.0/24 [110/2] via 10.0.12.2, 04:15:05, GigabitEthernet0/0
```

```
D 10.10.40.0/24 [90/28416] via 10.0.13.2, 00:34:54, GigabitEthernet0/1
```

```
O 10.10.50.0/24 [110/2] via 10.0.12.2, 04:15:05, GigabitEthernet0/0
```

```
D 10.10.60.0/24 [90/28416] via 10.0.13.2, 00:34:54, GigabitEthernet0/1
```

4. Switching and VLANs

4.1 VLAN Configuration

Six VLANs were created to segment the network by department and function. Each VLAN isolates broadcast domains, improving security and performance. VLAN 99 serves as the native VLAN for trunk links to prevent VLAN hopping attacks. Hosts are assigned to appropriate VLANs via access port configuration with port security enabled.

```
SW-DIST-A#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Pol, Po2, Fa0/2, Fa0/3 Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/22, Fa0/24, Gig0/2
10 Management	active	
20 IT_Dept	active	
30 HR_Dept	active	
40 Finance	active	
50 Servers	active	
60 Sales	active	
99 Native_Unused	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

4.2 VTP Configuration

VTP (VLAN Trunking Protocol) was configured with SW-DIST-A as the VTP Server for domain "ENTERPRISE". Site A access switches operate in VTP Client mode, receiving VLAN information automatically. Site B switches use VTP Transparent mode because VTP cannot propagate through routers, requiring local VLAN configuration at the branch site.

Switch	VTP Mode	Domain	Site
SW-DIST-A	Server	ENTERPRISE	Site A (HQ)
SW-ACC1/ACC2/SRV	Client	ENTERPRISE	Site A (HQ)
SW-DIST-B/ACC3/ACC4	Transparent	ENTERPRISE	Site B (Branch)

```
SW-DIST-A#show vtp status
VTP Version capable : 1 to 2
VTP version running : 1
VTP Domain Name : ENTERPRISE
VTP Pruning Mode : Disabled
VTP Traps Generation : Disabled
Device ID : 0002.17E9.A300
Configuration last modified by 0.0.0.0 at 3-1-93 00:35:03
Local updater ID is 10.10.10.11 on interface Vll10 (lowest numbered VLAN interface found)

Feature VLAN :
-----
VTP Operating Mode : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 12
Configuration Revision : 14
MD5 digest : 0x6F 0x4C 0x5D 0xAE 0x52 0x4E 0x09 0xC9
             0x9F 0x00 0x26 0x46 0x34 0x18 0x27 0xC0

SW-DIST-A#
```

```
SW-ACC1#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : ENTERPRISE
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                 : 0060.7074.1750
Configuration last modified by 0.0.0.0 at 3-1-93 00:35:03
```

Feature VLAN :

```
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 12
Configuration Revision   : 14
```

4.3 Inter-VLAN Routing

Router-on-a-stick configuration enables inter-VLAN routing using subinterfaces on R-SITE-A (G0/1) and R-SITE-B (G0/1). Each subinterface is assigned to a specific VLAN using 802.1Q encapsulation and serves as the default gateway for that VLAN. This approach is cost-effective and centralizes routing decisions on the router.

4.4 STP Configuration

Rapid PVST+ was implemented across all switches for fast convergence. SW-DIST-A is configured as the Root Primary for all VLANs, ensuring a predictable spanning tree topology. Access ports are configured with PortFast for immediate edge port activation and BPDU Guard on Finance switches for enhanced security.

```
SW-DIST-A#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: default Management IT_Dept HR_Dept Finance Servers Sales Native_Unused
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast             is disabled
Configured Pathcost method used is short
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	3	0	0	1	4
VLAN0010	1	0	0	3	4
VLAN0020	1	0	0	3	4
VLAN0030	2	0	0	2	4
VLAN0040	2	0	0	2	4
VLAN0050	2	0	0	2	4
VLAN0060	2	0	0	2	4
VLAN0099	1	0	0	3	4
8 vlans	14	0	0	18	32

5. DHCP Integration

5.1 DHCP Configuration

DHCP pools were configured on R-SITE-A for Site A VLANs (IT, HR, Servers) and on R-SITE-B for Site B VLANs (Finance, Sales). Each pool provides automatic IPv4 addressing with appropriate default gateway and DNS server settings. Address exclusions reserve the first 20 addresses for static assignment to servers and network devices.

Pool Name	Network	Gateway	DNS Server
VLAN20_IT	10.10.20.0/24	10.10.20.1	10.10.50.10
VLAN30_HR	10.10.30.0/24	10.10.30.1	10.10.50.10
VLAN40_FIN	10.10.40.0/24	10.10.40.1	10.10.50.10
VLAN60_SALES	10.10.60.0/24	10.10.60.1	10.10.50.10

```
R-SITE-A#show ip dhcp binding
```

IP address	Client-ID/ Hardware address	Lease expiration	Type
10.10.20.21	0002.1643.3C52	--	Automatic
10.10.20.22	0001.C7A8.A047	--	Automatic
10.10.20.23	000C.8554.15CA	--	Automatic
10.10.30.21	0010.1160.05C6	--	Automatic
10.10.30.22	0001.C903.B8C8	--	Automatic
10.10.30.23	0010.11E8.10DB	--	Automatic

```
R-SITE-B#show ip dhcp binding
```

IP address	Client-ID/ Hardware address	Lease expiration	Type
10.10.40.21	0060.704A.4999	--	Automatic
10.10.40.22	000C.855A.CC5C	--	Automatic
10.10.40.23	0010.1191.2ED0	--	Automatic
10.10.60.21	0060.7067.D637	--	Automatic
10.10.60.22	00E0.F92A.B570	--	Automatic
10.10.60.23	00D0.58A6.81AA	--	Automatic

6. Testing and Verification

6.1 Connectivity Tests

Comprehensive ping tests verified end-to-end connectivity across all VLANs and sites:

Test	Command	Result
IT to HR (Inter-VLAN)	ping 10.10.30.21	PASS
IT to Finance (Cross-Site)	ping 10.10.40.21	PASS
IT to DNS Server	ping 10.10.50.10	PASS
IT to Sales (Cross-Site)	ping 10.10.60.21	PASS
DNS Resolution	nslookup www.enterprise.local	PASS

```
C:\>ping 10.10.30.21

Pinging 10.10.30.21 with 32 bytes of data:

Reply from 10.10.30.21: bytes=32 time=30ms TTL=127
Reply from 10.10.30.21: bytes=32 time=8ms TTL=127
Reply from 10.10.30.21: bytes=32 time<1ms TTL=127
Reply from 10.10.30.21: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.30.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 30ms, Average = 9ms

C:\>ping 10.10.50.10

Pinging 10.10.50.10 with 32 bytes of data:

Reply from 10.10.50.10: bytes=32 time<1ms TTL=127
Reply from 10.10.50.10: bytes=32 time<1ms TTL=127
Reply from 10.10.50.10: bytes=32 time=13ms TTL=127
Reply from 10.10.50.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>nslookup www.enterprise.local

Server: [10.10.50.10]
Address: 10.10.50.10

Non-authoritative answer:
Name: www.enterprise.local
Address: 10.10.50.20
```


7. Reflection and Lessons Learned

7.1 Challenges Faced

VTP Propagation Across Sites:

Initially, Site B switches were configured as VTP clients, but VLAN creation failed with "VTP VLAN configuration not allowed when device is in CLIENT mode." The root cause was that VTP cannot propagate through routers, only through Layer 2 trunk links. The solution was to configure Site B switches in VTP Transparent mode, allowing local VLAN creation while still participating in the VTP domain.

Trunk Link Configuration:

DHCP failed for several VLANs despite correct VLAN configuration. Investigation using "show interfaces trunk" revealed that trunk links between distribution and access switches were not formed properly. The 3560 switches required explicit "switchport trunk encapsulation dot1q" commands before trunk mode could be enabled, while 2960 switches did not support this command. Careful attention to switch model differences resolved the issue.

Access Port VLAN Assignment:

Some PCs showed DHCP timeout despite correct VLAN configuration on switches. The "show vlan brief" command revealed that access ports remained in VLAN 1 instead of their assigned VLANs. Re-applying the access VLAN configuration with explicit "switchport mode access" followed by "switchport access vlan X" commands resolved the issue.

7.2 Technical Insights Gained

1. **Layer-by-Layer Troubleshooting:** Always verify Layer 2 connectivity before troubleshooting Layer 3. Many routing issues stem from underlying switching problems.
2. **Protocol Domain Boundaries:** VTP, STP, and other Layer 2 protocols operate within broadcast domains. Routers create boundaries that these protocols cannot cross.
3. **Redistribution Complexity:** Route redistribution between OSPF and EIGRP requires careful metric configuration. Administrative distance differences affect which path is preferred.
4. **Switch Model Differences:** Layer 3 switches (3560) and Layer 2 switches (2960) have different command syntaxes. Always verify command compatibility before applying configurations.

7.3 Conclusion

This project successfully demonstrated enterprise network design principles including hierarchical topology, dual routing protocols, VLAN segmentation, and centralized DHCP services. The implementation of OSPF and EIGRP with mutual redistribution showcased real-world scenarios where organizations may need to integrate different routing domains. Key takeaways include the importance of systematic troubleshooting, understanding protocol boundaries, and verifying configurations at each network layer before proceeding to the next.

End of Document